

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{state} : Loc \times Loc \times Loc \Rightarrow State \\ \text{far} : Loc \\ \text{near} : Loc \\ \text{in} : Loc \\ \text{ctrl0} : Loc \\ \text{ctrl1} : Loc \\ \text{ctrl2} : Loc \\ \text{ctrl3} : Loc \\ \text{up} : Loc \\ \text{down} : Loc \\ \text{success} : State \\ \text{error} : State \end{array} \right\}$$

$$\left\{ \begin{array}{l} 1 : \quad \text{state}(\text{far}, \text{ctrl0}, g) \rightarrow \text{state}(\text{near}, \text{ctrl1}, g) \\ 2 : \quad \text{state}(\text{near}, ct, g) \rightarrow \text{state}(\text{in}, ct, g) \\ 3 : \quad \text{state}(\text{in}, \text{ctrl2}, g) \rightarrow \text{state}(\text{far}, \text{ctrl3}, g) \\ 4 : \quad \text{state}(t, \text{ctrl1}, \text{up}) \rightarrow \text{state}(t, \text{ctrl2}, \text{down}) \\ 5 : \quad \text{state}(t, \text{ctrl3}, \text{down}) \rightarrow \text{state}(t, \text{ctrl0}, \text{up}) \\ 6 : \quad \text{state}(t, ct, g) \rightarrow \text{success} \\ 7 : \quad \text{state}(\text{in}, ct, \text{up}) \rightarrow \text{error} \end{array} \right\}$$